@henryleberre **Henry A. LE BERRE /ɑ̃ʁiləbɛʁ/ [U.S. & French Citizen]** hberre3@gatech.edu

WORK EXPERIENCE

- **Space Exploration Technologies | Software Engineering Intern** — Los Angeles, California, USA – Fall 2023
  Raptor Combustion Devices Engineering | Raptor Production Group

25% speed up of the **Raptor Combustion CFD Code** on production simulations through C++ CUDA GPU kernel optimizations.

- **Verkada | Software Engineering Intern** — San Mateo, California, USA – Summer 2023
  Device Platform Team | Alarms and Intrusion
- **Computational Physics Group | Student Assistant** — Atlanta, Georgia, USA – Summer 2022 & 2024

EDUCATION

- **Georgia Institute of Technology** (Atlanta, GA, USA) | Bachelor's in Science in **Computer Science** — Summer 2021 – Fall 2024
  GPA: 3.83 | Concentration: **Modeling & Simulation** and **Systems & Architecture** with the **Research Option**.
- **Lycée Lavoisier** (High School in Paris Vè, France) — September 2018 – July 2021

PUBLICATIONS

- **Method for portable, scalable, and performant GPU-accelerated simulation of multiphase compressible flow** — May 2024
  Computer Physics Communications — 10.1016/j.cpc.2024.109238
- **Early Application Experiences on a Modern GPU-Accelerated Arm-based HPC Platform** — September 2022
  HPC Asia '23. International Workshop on Arm-based HPC: Practice and Experience (IWAHPCE) — 10.1145/3581576.3581621

AWARDS

- **Verkadathon** (Verkada Hackathon) **Winner** for "Best Use of Verkada Products" with Viraj Ramakrishnan and Jay Chou — Summer 2023
- **President's Undergraduate Research Award** (**PURA**) **Travel** from the **Georgia Institute of Technology** — Fall 2022
- **Faculty Honors** and **Dean's List** from the **Georgia Institute of Technology**
- The State of Georgia's **HOPE Scholarship**.
- **National General Baccalaureate** (« Baccalauréat Général ») with **Highest Honors** (Mention « très bien ») — 2021
- **Highest Honors** (« Félicitations du conseil de classe ») for 9 consecutive trimesters — 2018 – 2021
- **Diplôme National du Brevet** with **Highest Honors** (Mention « très bien ») — 2018

ACADEMIC RESEARCH AFFILIATIONS

Computational Physics Group (CPG) at Georgia Tech | Multi-Component Flow Code (MFC) — November 2021 – December 2024

- Implementing reactive flow modeling (combustion).
- Offloaded and optimized the Fortran/MPI flow solver to efficiently run and scale simulations on leadership-class distributed systems of GPUs such as OLCF Summit, with PhD student Anand Radhakrishnan.
- Discovered and reported bugs in GNU and HPE Cray offloading compilers for Fortran, to support running on OLCF Frontier.
- Modernized the codebase, adding a test suite, CI, CMake support, and a custom toolchain for case-specific compile-time optimizations.

Space Systems Design Lab (SSDL) at Georgia Tech | GT-II Satellite — November 2021 – Dec. 2022

Designed & implemented the GT-II satellite's Over-The-Air update (OTA) protocols, on bare-metal, from ground systems to in-orbit handling.

LEADERSHIP, SERVICE, AND MISCELANIOUS

- **Student Volunteer** at **Supercomputing 2023 & 2024** | Denver, Colorado, USA & Atlanta, GA, USA — November 11-17 2023 & 2024
- **Mentor** for team CEREAL at the **Georgia Tech Open Hackathon** (C++/MPI/Cuda/NVSHMEM) — January 2023
- **Co-Chair** of the **Georgia Tech French Club** — August 2022 – Fall 2024
- **Languages: English (Native), French (Native), and German (High School Level)**

TALKS, POSTERS, AND PRESENTATIONS

- **Compressible multi-species flow simulation on OLCF Frontier via OpenACC** — March 7, 2024
  American Physical Society (March 2024), Minneapolis, Minnesota, USA — Abstract
- **Fast simulation of multiphase compressible flows through GPU acceleration** — April 5, 2023
  A stochastic computational method for bubbly flows with first steps towards representing inception — April 5, 2023
  11th International Conference on Multiphase Flow (ICMF). Kobe, Japan — Program
- **Compressible multiphase flow simulation at near-exascale via a scalable GPU implementation** — March 7, 2023
  American Physical Society (March 2023), Las Vegas, Nevada, USA — Abstract
- **Towards exascale multiphase compressible flow simulation via scalable interface capturing-based solvers…** — November 21, 2022
  American Physical Society (November 2022), Indianapolis, Indiana, USA — Abstract
- **Scalable GPU Accelerated Simulation of Multiphase Compressible Flow** — November 15, 2022
  Supercomputing 2022 (SC'22), Dallas, Texas, USA — Poster | Paper

- **Exascale for humans: A development environment for MFC** February 19, 2022
  Georgia Scientific Computing Symposium 2022 (GSCS 2022), Atlanta, Georgia, USA Website

Last updated: August 7, 2024.